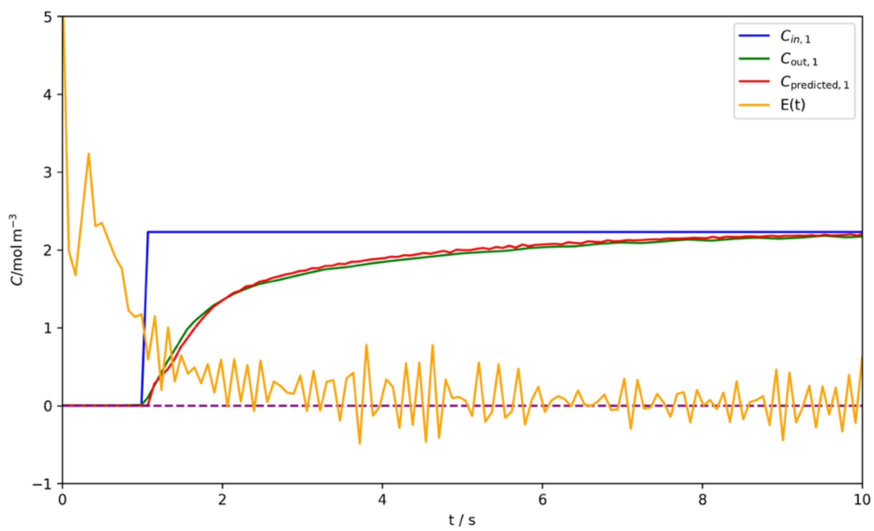
Supporting Material – The Residence Time Distribution Analysis via Convolutional Neural Networks: A Promising Approach

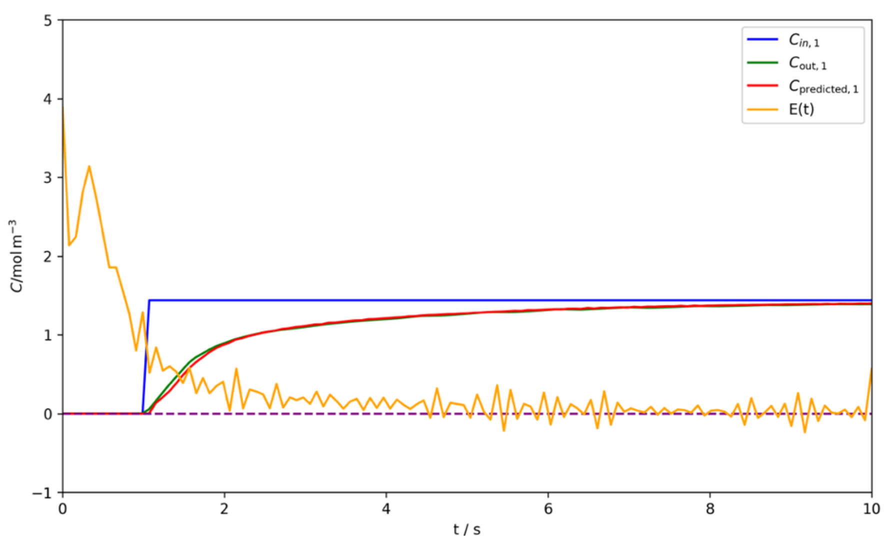
Tuana Oyuncu

The general responses of the model for the first measurement point with varied concentrations are illustrated in the Figure S1.

(a)



(b)



(c)

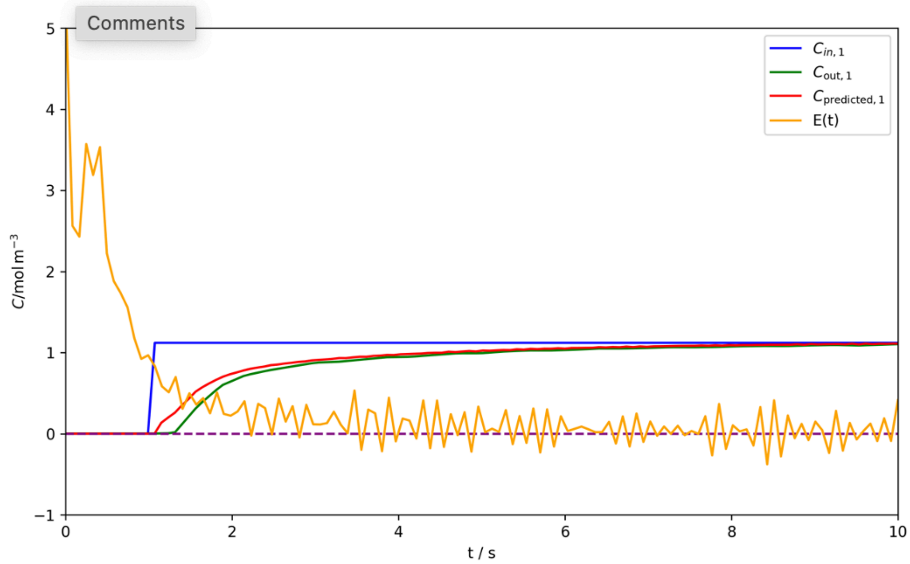
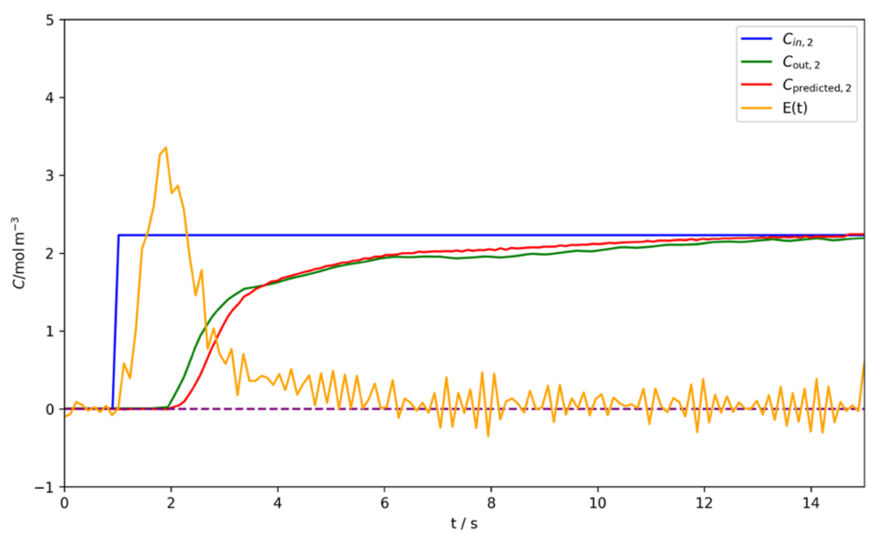


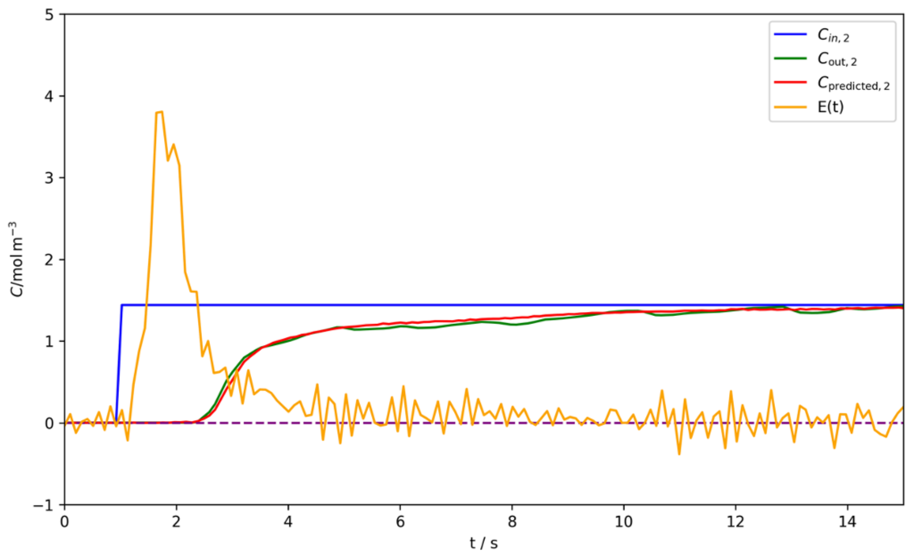
Figure S1. The general responses of the model for the first measurement point with varied concentrations

The general responses of the model for the second measurement point with varied concentrations are illustrated in the Figure S2.

(a)



(b)



(c)

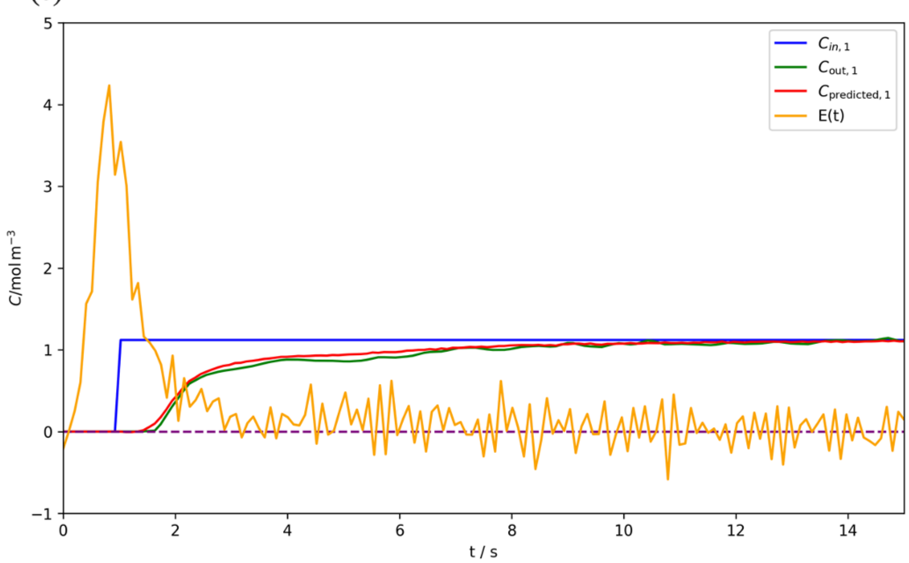
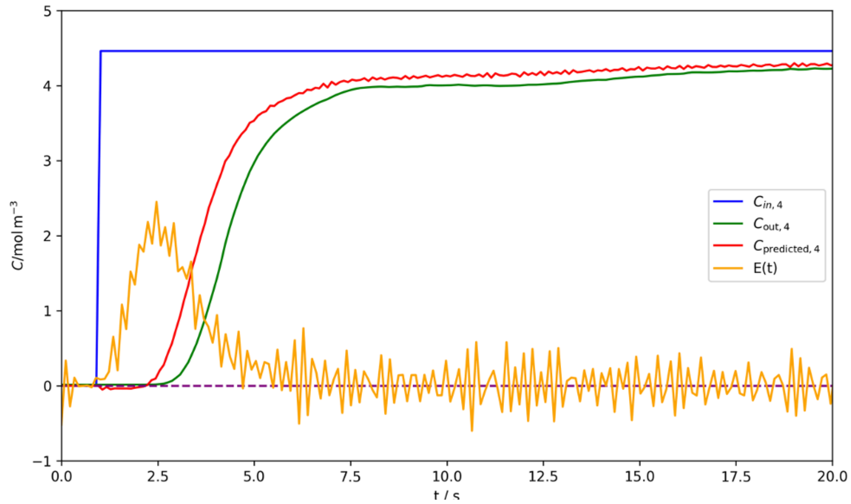


Figure S2. The general responses of the model for the second measurement point with varied concentrations

The general responses of the model for the third measurement point with varied concentrations are illustrated in the Figure S3.

(a)



(b)

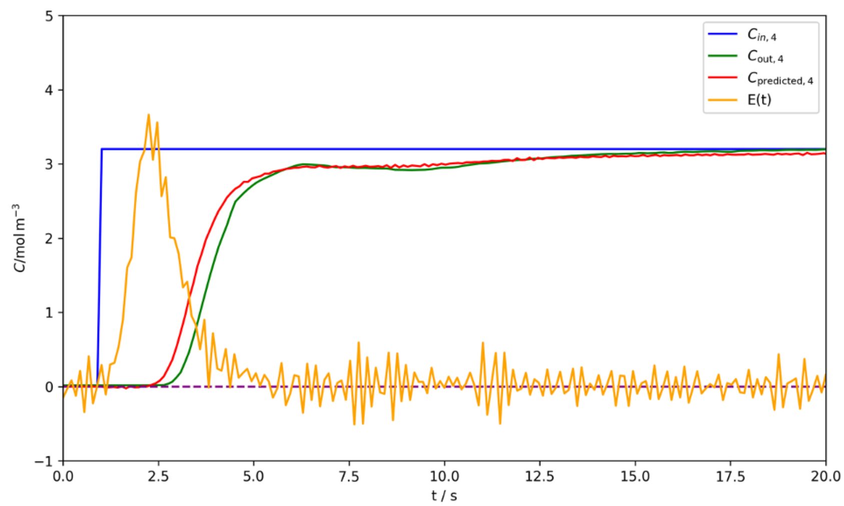


Figure S3. The general responses of the model for the third measurement point with varied concentrations

The responses of the model according to varied learning rate during the training are illustrated in Figure S4. The figure shows the instability during the training.

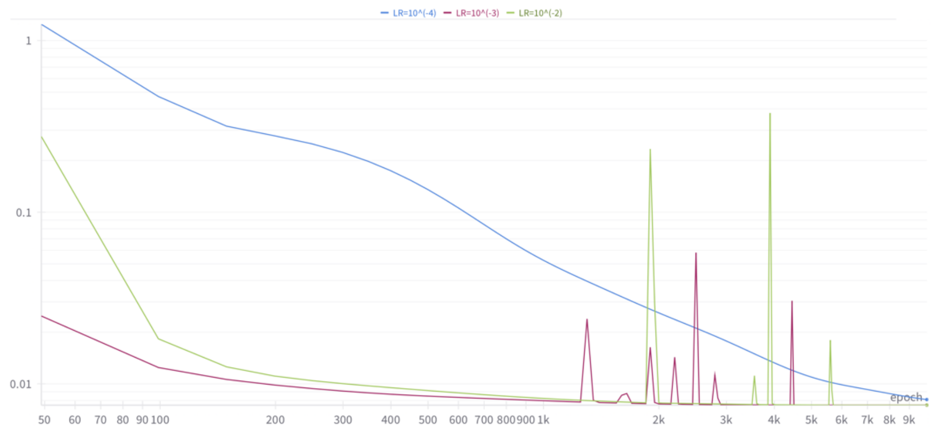


Figure S4. The instability through the epoch number

*LR= Learning Rate

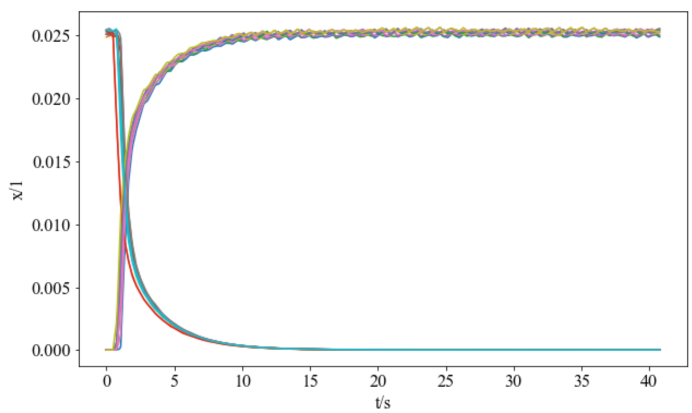


Figure S5. The experimental data of 200

Table S1. The calculated model parameters for the position (1)

n_{o}	500
n_{i}	377
n_{E}	122
Δt_o	41 s
Δt_{i}	31 s
Δt_{E}	10 s

Table S2. The calculated model parameters for the position (2)

n_{o}	500
n_{i}	336
n_{E}	163
Δt_o	46 s
Δt_{i}	31 s
Δt_{E}	15 s

Table S3. The calculated model parameters for the position (4)

n_{o}	500
n_{i}	321
n_{E}	178
Δt_o	56 s
Δt_{i}	20 s
Δt_{E}	36 s

Table S4. The Flow Rate, 100 mL/min

Helium	10 mL/min
Hydrogen	10 mL/min
Argon	80 mL/min

Table S5. The Flow Rate, 150 mL/min

Helium	10 mL/min
Hydrogen	10 mL/min
Argon	130 mL/min

Table S6. The Flow Rate, 200 mL/min

Helium	10 mL/min
Hydrogen	10 mL/min
Argon	180 mL/min